

PitchClipers: Learning to Select Context-Complete Football Highlights

Ahmed Gabr

Iyad Khamis

Zaid Rafi

Abstract

Accurate action timestamps do not guarantee useful football highlights: a clip must show sufficient build-up and outcome while avoiding unrelated footage. We propose a lightweight, learnable clip-selection layer over pretrained action spotting, augmented by our Football Analytics module. The selector estimates contextual sufficiency from event confidence, candidate boundaries, and tracking evidence. We define a detector-independent Context Utility score combining event support, contextual completeness, and temporal precision. Our research question is whether learned clip selection improves highlight utility over confidence filtering under a fixed viewing budget. The study combines automatic evaluation against shared annotations with human assessment and an interactive demonstration.

1. Problem and Contribution

SoccerNet-v2 [1] provides action timestamps rather than highlight intervals. We define a *useful highlight* as a clip that contains a correctly identified target event, preserves its necessary build-up and outcome, and limits unrelated footage. Spotting accuracy alone cannot establish these properties.

Our intended contributions are a low-cost clip-quality predictor, a detector-independent metric, and an ablation of player–ball evidence. The visual backbone remains frozen to limit compute.

2. Proposed Computer-Vision Method

Event proposals. Our video-to-spot pipeline uses frozen ResNet-152 descriptors sampled at 2 FPS, the released 512-dimensional PCA representation, and the pretrained CALF model [2]. Its temporal context model produces proposals (t, c, s) for 17 football actions. The clip-selection interface also accepts outputs from other spotters, allowing the proposed method to be evaluated independently of CALF.

Learnable clip selection. Around each proposal, we enumerate a small bank of asymmetric windows, spanning 4–20 seconds before and 3–15 seconds after the event. For candidate I , features z_I include class, confidence, window lengths, nearby proposals, ball visibility, and player–ball

association changes. A regularized logistic scorer predicts contextual sufficiency:

$$q_\theta(I) = \sigma(\theta^\top z_I), \quad \mathcal{L} = \text{BCE}(q_\theta, y_I) + \lambda \|\theta\|_2^2. \quad (1)$$

Here y_I is a human judgment that the clip contains the event and enough context to understand it. Training uses development matches; thresholds and regularization are selected on separate validation matches. All candidates from a match remain in one split. At inference, candidates are ranked by predicted quality with a duration penalty; overlaps are merged and final selections obey a fixed viewing budget. This constitutes a learned selection module beyond per-class threshold tuning.

Our Football Analytics module. Player, referee and ball detection, ByteTrack association, jersey-color team assignment, and proximity-based ball-control estimation provide spatial evidence. Missing ball observations receive explicit missingness indicators rather than implied possession. Player positions additionally support an offside visualization cue; validating offside requires calibrated pitch geometry and the ball-contact instant. We will test whether these analytics improve context selection, rather than assume tracking accuracy guarantees better highlights.

3. Highlight Metric and Evaluation

For each target event j , annotators specify a sufficient-context interval g_j and its class and timestamp. Let $\mathcal{C}$ be exported clips, U their temporal union, and $G = \bigcup_j g_j$. Define contextual coverage R , temporal precision P , and supported-clip fraction S :

$$R = \frac{1}{N} \sum_j \max_{I \in \mathcal{C}} \frac{|I \cap g_j|}{|g_j|}, \quad P = \frac{|U \cap G|}{|U|}. \quad (2)$$

A clip is supported if it contains an annotated event with a matching predicted class. Define *Context Utility* as $H = S2PR/(P+R)$, with zero for empty output or zero denominator. Component scores reveal false clips, excess footage and truncated context. We also report complete-context rate and repeated footage. Comparisons use the same viewing budget.

Once reference intervals are annotated, this calculation is automatic for any action spotter producing labeled clips.

SoccerNet timestamps alone cannot provide true context intervals. A point-label-only proxy using fixed reference windows can be computed without new annotation, but must be reported separately because it encodes assumed context rather than human utility.

4. Study Plan and Expected Demonstration

The planned ablation compares global thresholds/fixed windows, class thresholds, learned selection using event features, and selection with Football Analytics features. Disjoint held-out matches are assessed with Context Utility, spotting mAP and runtime. Two reviewers independently annotate context and rate sufficiency; agreement and metric–rating correlation assess validity. The initial study uses a small annotated subset, with rare-class findings treated as exploratory.

The interactive demo will show event timelines, player–ball evidence, adjustable context and selected clips with ordered MP4 export. Evaluation will test whether learned selection improves completeness and concision.

References

- [1] A. Delière et al. SoccerNet-v2: A Dataset and Benchmarks for Holistic Understanding of Broadcast Soccer Videos. CVPRW, 2021.
- [2] A. Cioppa et al. A Context-Aware Loss Function for Action Spotting in Soccer Videos. CVPR, 2020.